

Functional Requirements:

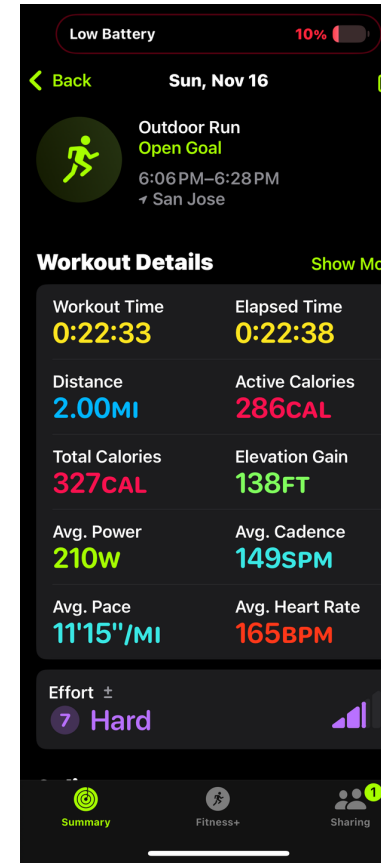
1. Allow user to start a specific workout type
2. Once a workout starts, display workout progress and a summary of stats in real-time:
 1. Elapsed Time
 2. Current Active Calories
 3. Current Heart Rate
 4. Distance so far in (Feet/miles)
 5. Average Pace
3. Allow user to end a workout
4. Upon workout end Apple Watch should send workout data as a report to a companion app. To the watch the companion app serves as a database/backend for all workouts storing each workout and associating to a calendar.
5. App remembers workout data for a single day and thats it, after that resets.

Non-Functional Requirements:

1. The app needs to integrate data from various data sources and services and update the UI thread quickly and seamlessly.
2. The app will need to seamlessly switch from being able to run in foreground and background. When the watch app comes back to the foreground resumption of the workout should feel instantaneous to the user.
3. Because its a apple watch app we are in a memory constrained environment (both in-memory and disk space), so optimize for memory along with quick main-thread updates.
4. Optimize for battery, don't want to drain battery quickly.

Assumptions and Non-Goals:

1. Currently assuming a very simple watch to phone/companion handoff when workout is done uploading stats. I noticed this handoff happens more frequently if you are on a run and bring your phone with you. Everytime you open the phone activity app it will get the latest activity/health data snapshot from the watch. Descoping for now.
2. We are not concerned about "move goal" and health ring visualizations and calculations.



- distance -> need location updates and GPS
- heartRate, power, and stepCount come from HealthKitSession
- elevationGain -> will come from location updates via GPS. A CLLocation object contains lat, long, and altitude.

